The Signal

A systematic washout-reversion overlay on the S&P 500, built on four causal anomaly detectors and an append-only audit trail.

INVESTMENT PROSPECTUS & PROPOSAL FOR CORPORATE CAPITAL

“The noise is what they sell you. The signal is what we extract.”

PREPARED FOR	The Office of the Chief Executive	PREPARED BY	Jacob E. Thomas, PhD · Results Generation
DOCUMENT DATE	2 September 2026	DATA AS OF	Close of 1 September 2026
MODEL VERSION	`2.0.0` · causal regime · commit `6c66783`	RECORD	1 November 2024 – 1 September 2026 (458 sessions)

IMPORTANT INFORMATION

Notice to the reader

Purpose. This document describes an internal, systematic equity strategy ("The Signal") and proposes a framework under which company capital could be allocated to it. It is prepared for the Office of the Chief Executive and the board or committee responsible for treasury policy. It is not an offer to sell, or a solicitation of an offer to buy, any security, and it is not investment, legal, tax or accounting advice. Any allocation decision remains solely with the company's governing bodies.

Hypothetical performance. The performance presented in Sections 5 to 8 is a *walk-forward simulation*: the detection engine is causal (every verdict for a trading day uses only data available at that day's close, a property enforced by the system's test suite), and the simulation fills every trade at the close of the session *after* the signal existed, after transaction costs of 5 basis points per side. It is nonetheless a simulation. It does not reflect actual trading, market impact, financing, taxes, or the behaviour of a human operator under stress. Hypothetical results have inherent limitations, including that they are generally prepared with the benefit of hindsight in the choice of universe and parameters. **Past performance, simulated or real, is not a reliable indicator of future results.**

Live record. Verdicts produced by scheduled runs on unseen data since 12 June 2026 are reported separately as the out-of-sample record (Section 6). That record is short and should be weighed accordingly.

Sources. Every figure in this document is computed from files committed to the strategy's repository as of the run dated 2026-09-01 (commit `6c66783`): the append-only signal ledger, the run history, the run-health report and the walk-forward backtest rendered to the operational dashboard. The document is regenerated by `reports/prospectus/build_prospectus.py`; no number in it is typed by hand. Price data are sourced from Yahoo Finance via the `yfinance` library and are subject to vendor revision.

Risk. The strategy is fully invested in US-listed equities at all times and carries the full market risk of the S&P 500 in addition to single-name risk. An investor could lose a substantial portion of the capital allocated. Section 9 sets out the principal risk factors and should be read in full before any decision is taken.

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1 Executive summary

The Signal is a long-only overlay on a passive S&P 500 position. The book holds SPY by default and moves 10% slices into individual large-cap securities only when the detection engine identifies a statistically extreme washout. It never shorts, never borrows, and never holds a derivative.

+50.5%
TOTAL RETURN
SPY alone +36.1% · 1.83 years

+14.3 pp
EXCESS OVER SPY
+$14,339 on $100,000

1.28
SHARPE RATIO
SPY 1.09 · beta 1.05

−18.6%
MAXIMUM DRAWDOWN
SPY −18.8% over the same window

The proposition

A security that closes more than 2.5 standard deviations below its own 20-day trend, on a move confirmed by at least two of four independent detection methods and larger than 1.0% in absolute terms, has historically been a washout rather than the start of a decline. The Signal buys those washouts with capital drawn from SPY, and returns the capital to SPY when price recovers to the trend line that was frozen at detection, when a bearish signal fires on the same name, or after 30 trading sessions, whichever comes first. Because idle capital is never cash, the strategy's return is the index return plus whatever the overlay adds or subtracts, and nothing else.

The record

Over the walk-forward record from 1 November 2024 to 1 September 2026, a $100,000 book following the Signal grew to **$150,482** (+50.5%) against **$136,143** (+36.1%) for the same capital left entirely in SPY. The overlay added +$14,339. It did so with a beta of 1.05 to the index, a correlation of 0.93, and a maximum drawdown (−18.6%) fractionally shallower than the index's own (−18.8%) through the same episode. The overlay funded 72 round trips in 45 names; 63% of the 70 closed trades were profitable, the average winner (+$770) exceeded the average loser (−$660), and the profit factor was 1.97.

Since the system went live on 12 June 2026, every verdict has been produced by a scheduled run on data the model had never seen. The live book has funded 7 trades, 6 of them profitable, for +$3,920 (Section 6). That record is encouraging and short; it is presented separately for that reason.

Why this is different from most systematic proposals

- **The backtest is honest by construction.** Every score for a trading day uses only data through that day; the test suite fails if appending future data changes any past verdict. The historical backfill is therefore the same machine as live operation, advanced one bar at a time.
- **The benchmark is the strategy's own idle asset.** Strategy and benchmark start from the same dollar on the same day; the gap between them is exactly the overlay's contribution.
- **The record cannot be quietly rewritten.** Verdicts are appended to a ledger committed to version control on every run; 642 signals and 1,944 anomaly verdicts are on the record today, each with its provenance.
- **The rules were fixed before the live record began.** Thresholds are sigma cut-offs, never percentiles of the run's own output; the entry set (washouts only) was chosen by a split-half study documented in Section 10.

> PROPOSED ACTION
>
> **Fund a segregated pilot book of $250,000** (ten slices of $25,000), mirroring the validated structure at 2.5× scale, subject to the conditions precedent in Section 11. Review at 90 days against the published guardrails. Scale to $500,000 and then $1,000,000 only on passing each review gate. At every tier the book remains fully invested in SPY except for the slices the Signal moves, so the company's market exposure is the same as holding the index; what is being underwritten is the overlay.

2 Investment thesis

Short-term overreaction in large caps is persistent, and it is measurable

The tendency of a security to rebound after an abrupt, statistically extreme decline is one of the oldest documented regularities in equity markets: the short-term reversal effect described by De Bondt and Thaler (1985), Jegadeesh (1990) and Lehmann (1990) among others. Forced selling, index and factor rebalancing, and the mechanical de-risking of leveraged holders push prices past the point that new information justifies; liquidity providers are compensated for absorbing that flow.

The effect was measured directly on this universe before any rule was set. Bars stretched more than roughly 3.5 standard deviations *below* trend rebounded +2.3% on average (+3.3% median) over the following ten sessions, while equally extreme stretches *above* trend drifted approximately nowhere. That asymmetry is the whole thesis: the Signal buys capitulations and treats blow-offs as exit information rather than as something to short. Across the 132 washout signals on the record, the average entry was -4.2 standard deviations and −11.0% below trend.

The default position is the market

Most systematic strategies hold cash between trades and are judged on absolute return. The Signal holds SPY between trades. That choice does three things. It means the company is never paying an opportunity cost for the overlay’s patience; over the record, +$33,481 of the +$50,482 gain came from the baseline itself. It makes the benchmark comparison exact rather than a matter of convention. And it bounds the question the board has to answer: not “should we own equities?” but “does this overlay add to the equities we would own anyway?”

The evidence is causal, separated by provenance, and audited

A backtest is only as good as its leakage discipline. Here the discipline is a property of the code rather than of the analyst: every detector standardises against its own trailing history, the pattern-matching method uses a left-only matrix profile that cannot see forward, the machine-learning component is refit walk-forward on blocks it never scores, and the causality invariant is executed by the test suite before every scheduled run. Signals produced on new bars by scheduled runs are tagged *live* and reported separately from the walk-forward history. Every verdict is an appended line in a version-controlled ledger; changing history would be visible in the commit log.

What the company is being asked to underwrite

Not a view on the market, and not a black box. The proposal is to let a documented, tested, and independently auditable rule move a bounded fraction of an index position into oversold large caps for a few weeks at a time, and to judge it against the index it draws from. The relevant question for the pilot is whether the live record continues to look like the walk-forward record; Section 11 sets out how that will be measured and what happens if it does not.

3 Strategy and process

Four independent detectors, one scale

Each trading day, four methods score every security in the universe. Each answers a different question, and each is standardised against the security's own trailing history so that a 3-sigma event means the same thing for a Treasury-bill fund as for a semiconductor stock. Scores share one fixed scale: 0.5 is two standard deviations, 0.625 is two and a half, 1.0 is four or more. Detectors observe 60 bars plus 20 standardisation bars before they may flag anything.

Fourier transform

Has the rhythm changed?

Divergence between the frequency spectrum of the trailing 60 bars and the expanding baseline up to the same day. Weight 20%.

Matrix profile

Has this ever happened before?

Distance from the latest 10-day pattern to its nearest neighbour strictly in the past (STUMPY left profile). Weight 25%.

Statistical ensemble

Do independent tests agree?

Return z-score (40%), one-sided seasonal residual (30%), walk-forward isolation forest (30%). Weight 30%.

EWMA trend

Is momentum abnormal for this name?

Deviation from the 20-day exponential average, divided by that deviation's own trailing volatility. Weight 25%.

From anomaly to signal

A day is anomalous when two or more methods flag it, or when the weighted consensus exceeds the sensitivity cut-off (medium: 0.625, the 2.5-sigma equivalent). The signal that follows depends on how far price sits from trend in units of its own volatility (`dev_z`) and on what the stretch is doing. A trade call additionally requires two methods in agreement and a move of at least 1.0% in absolute terms, so a genuinely unusual but economically trivial wiggle in a T-bill fund stays informational.

Buy — stretch at or beyond −2.5σ and either climactic (breakout) or already fading. *The only signal that invests capital.*

Sell — stretch at or beyond +2.5σ, climactic or fading. Divests any slice held in the name.

Long / Short — stretch beyond ±1.5σ and still building. Informational momentum flags; Short also divests.

Reduce / Watch — structural regime change (Fourier and matrix profile agree), or an anomaly with no tradable set-up. Informational.

The capital model

The backtest, and the proposed live operation, follow one conserved pool of capital; nothing is notional.

1. **Baseline.** The book starts 100% in SPY. Every dollar is either in SPY or in a signal position.
2. **Invest.** A Buy signal moves one slice ($10,000 on a $100,000 book, i.e. 10%) from SPY into the security at the close of the first session *strictly after* the signal existed. If the baseline holds less than half a slice, the signal is skipped and counted; capital is never invented.
3. **Divest.** The slice returns to SPY at the earliest of: the close touching the trend value frozen at detection; the first Sell or Short signal on the name; the 30-session time stop. Positions still open are marked at the last close.
4. **Costs.** 5 basis points per side on every stock transaction. The SPY leg is treated as frictionless.
5. **Benchmark.** The identical capital left 100% in SPY on the same calendar.

PROTOCOL PARAMETERS

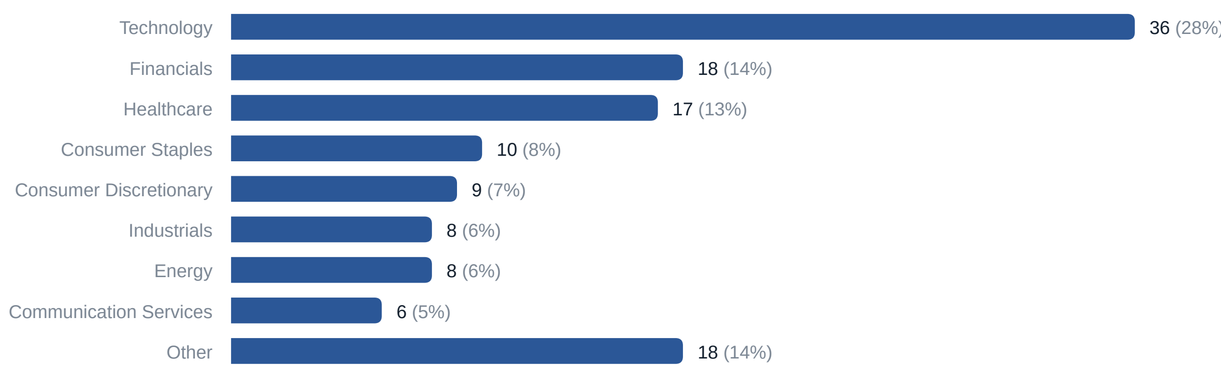
Anchor date	2024-11-01 (fixed; detection and backtest run forward from here)
Sensitivity	medium — reversion threshold 2.5σ, momentum threshold 1.5σ
Materiality gate	|deviation from trend| ≥ 1.0%
Entry set	BUY only (LONG informational; fixed by the split-half study, Section 10)
Slice / capital	$10,000 / $100,000 (10%)
Time stop	30 trading sessions
Transaction cost	5 bps per side
Model version	`2.0.0` — part of every ledger row's primary key

4 Investment universe

130 securities in 12 categories: the full top 100 US-listed tickers by market capitalisation, the index and sector funds Wall Street sells, and a curated set of thematic names. All are liquid, US-listed, and priced daily.

The universe is a registry in code (`config.TICKER_REGISTRY`): ticker, display name, category, sector, fund flag. Adding a name triggers a walk-forward backfill of its history on the next run without touching any other security's record. 14 entries are funds, including three mutual funds (DVRUX, QGRPX, BNUEX) held for comparison; the operational implications of trading those are addressed in Section 9.

COMPOSITION BY SECTOR



Number of securities per sector, 130 in total. Technology is the largest single sector by construction of the mega-cap set; the Signal's per-name slice cap, not sector weights, governs exposure.

CATEGORIES

CATEGORY	NAMES	EXAMPLES
Engines of the Republic	15	`CAT, DE, HON, LMT, GE, WMT …`
Critical Choke Points	14	`NEE, D, KMI, WMB, V, MA …`
Big Tech	8	`AAPL, NVDA, META, TSLA, NFLX, CRM …`
Consumer Leaders	8	`PG, KO, PEP, MCD, NKE, SBUX …`
Financial Titans	6	`GS, MS, BAC, C, WFC, BLK`
Healthcare & Pharma	5	`LLY, ABBV, MRK, AMGN, BMY`
Energy & Commodities	3	`XOM, CVX, COP`
Wall Street's Darlings	10	`SPY, QQQ, DIA, IWM, VTI, VOO …`
The Mega-Cap 100	55	`TSM, AVGO, MU, ASML, INTC, CSCO …`
The Clergy House A Priest's bespoke collection.	2	`ASTS, STM`

CATEGORY	NAMES	EXAMPLES
Reserve	1	BIL
UBS Funds	3	DVRUX, QGRPX, BNUEX

The complete membership is listed in Appendix B.

5 Performance record

All figures are computed from the daily mark-to-market value of the $100,000 book (a real capital base, so daily returns are well defined) over 458 sessions from 1 November 2024 to 1 September 2026. The first signal could not fire until 4 February 2025, when the detectors had accumulated their warm-up history; before that date the book was identical to SPY, and the shaded band on the chart marks the period.

GROWTH OF $100,000

Daily portfolio value. Both series start from the same dollar on 2024-11-01; the gap at any date is the cumulative contribution of the overlay. The live record (Section 6) begins at the vertical marker.

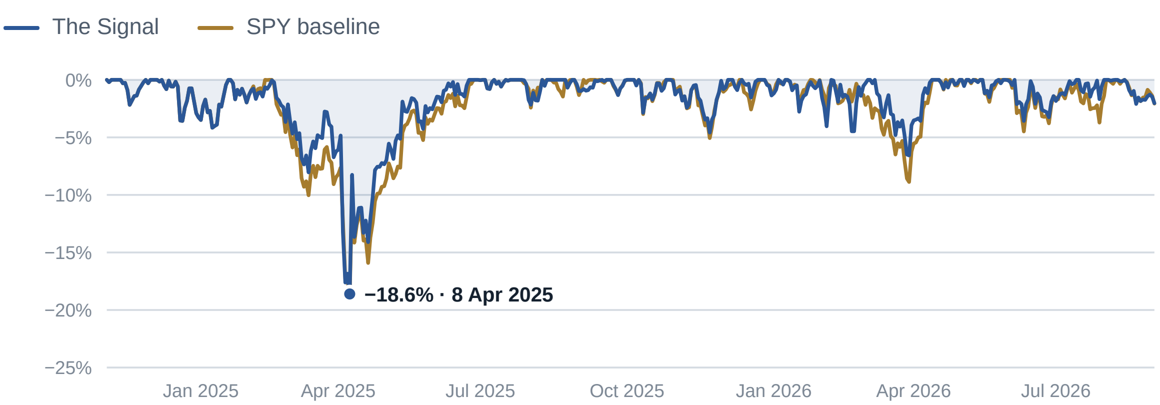
SUMMARY STATISTICS

FULL RECORD, 2024-11-01 – 2026-09-01	THE SIGNAL	SPY BASELINE	DIFFERENCE
Ending value of $100,000	$150,482	$136,143	+$14,339
Total return	+50.5%	+36.1%	+14.3 pp
Compound annual growth rate	25.0%	18.3%	
Annualised volatility	19.1%	17.0%	
Sharpe ratio (0% risk-free)	1.28	1.09	
Sortino ratio	1.92	1.63	
Maximum drawdown (peak to trough)	−18.6%	−18.8%	

FULL RECORD, 2024-11-01 – 2026-09-01	THE SIGNAL	SPY BASELINE	DIFFERENCE
Calmar ratio	1.34	0.98	
Best / worst day	+12.7% / −8.8%	+10.5% / −5.9%	
RELATIVE TO SPY			
Beta / correlation	1.05 / 0.93	1.00 / 1.00	
Annualised alpha (CAPM, 0% risk-free)	+5.0%		
Tracking error / information ratio	6.8% / 0.86		
Up-market / down-market capture	100% / 92%		
OVERLAY ACTIVE ONLY, 5 FEBRUARY 2025 – 1 SEPTEMBER 2026			
Total return	+41.7%	+28.2%	+13.5 pp
CAGR / Sharpe	24.9% / 1.22	17.2% / 0.99	
Tracking error / information ratio	7.4% / 0.93		

Sharpe and Sortino use daily returns annualised by √252 with a 0% risk-free rate, the convention used on the operational dashboard. Maximum drawdown is measured peak to trough on the daily series; see Appendix C on why this differs from the dashboard's drawdown tile.

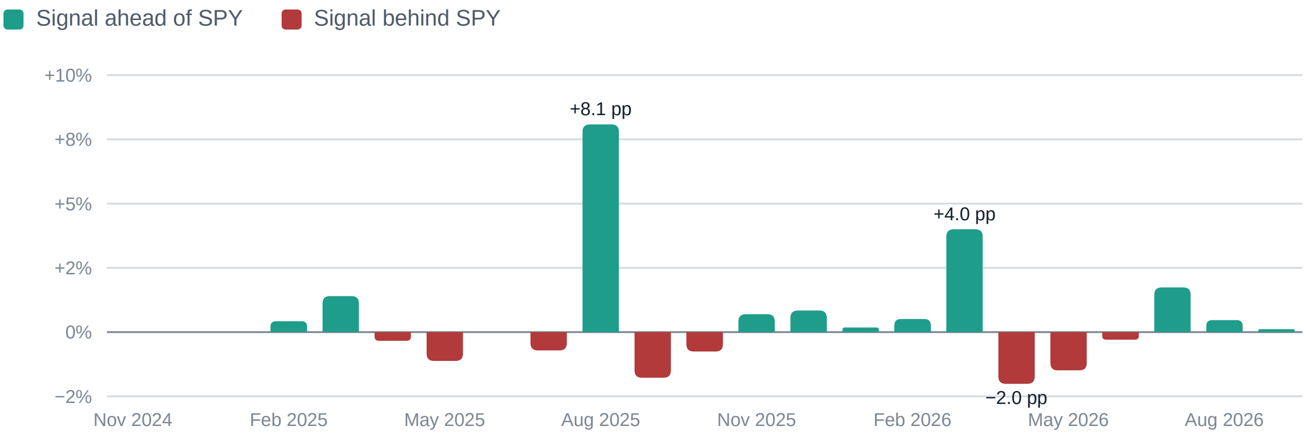
Drawdowns



Decline from the running peak. The Signal's deepest drawdown (−18.6%, −$19,940) ran from 23 Jan 2025 to 8 Apr 2025 and was recovered by 24 Jun 2025; SPY fell −18.8% through the same episode. The overlay neither deepened nor materially shortened the index's own drawdown.

The deepest drawdown on the record ran from 23 Jan 2025 to 8 Apr 2025. The book was fully deployed into washouts through that decline: the Mar 2025 entries were the worst monthly cohort on the record (−$5,284 on 9 slices) and the Apr 2025 entries the best (+$5,515 on 6 slices). The strategy recovered its peak 77 days after the trough, 2 sessions ahead of the index. The best single day for the book (+12.7%, 9 Apr 2025) and for the index were the same session.

Monthly and calendar-year returns



Monthly return of the Signal minus the monthly return of SPY, in percentage points. Of the 20 months with the overlay active, the Signal finished ahead in 12 and behind in 8. Best relative month Aug 2025 (+8.1 pp); worst Apr 2026 (−2.0 pp).

MONTH	SIGNAL	SPY	EXCESS
Nov 2024	+5.5%	+5.5%	—
Dec 2024	−2.4%	−2.4%	—
Jan 2025	+2.7%	+2.7%	—
Feb 2025	−0.8%	−1.3%	+0.4 pp
Mar 2025	−4.2%	−5.6%	+1.4 pp
Apr 2025	−1.2%	−0.9%	−0.3 pp
May 2025	+5.2%	+6.3%	−1.1 pp
Jun 2025	+5.1%	+5.1%	+0.0 pp
Jul 2025	+1.6%	+2.3%	−0.7 pp
Aug 2025	+10.1%	+2.1%	+8.1 pp
Sep 2025	+1.8%	+3.6%	−1.8 pp
Oct 2025	+1.6%	+2.4%	−0.8 pp

MONTH	SIGNAL	SPY	EXCESS
Nov 2025	+0.9%	+0.2%	+0.7 pp
Dec 2025	+0.9%	+0.1%	+0.8 pp
Jan 2026	+1.7%	+1.5%	+0.2 pp
Feb 2026	−0.4%	−0.9%	+0.5 pp
Mar 2026	−0.9%	−4.9%	+4.0 pp
Apr 2026	+8.5%	+10.5%	−2.0 pp
May 2026	+3.8%	+5.3%	−1.5 pp
Jun 2026	−1.3%	−1.0%	−0.3 pp
Jul 2026	+1.8%	+0.0%	+1.7 pp
Aug 2026	+3.1%	+2.7%	+0.5 pp
Sep 2026	−0.6%	−0.7%	+0.1 pp

CALENDAR YEAR	SPAN	SIGNAL	SPY	EXCESS
2024 (partial)	1 Nov 2024 – 31 Dec 2024	+3.0%	+3.0%	+0.0 pp
2025	2 Jan 2025 – 31 Dec 2025	+25.6%	+17.7%	+7.9 pp
2026 (partial)	2 Jan 2026 – 1 Sep 2026	+16.3%	+12.3%	+4.0 pp

Dimmed months precede the first funded trade; the book was identical to SPY.
Monthly figures are from month-end portfolio values.

6 The live record

Since 12 June 2026 the system has run on a schedule, scoring each new session once and freezing the verdict. Trades below were entered on signals the model produced without any knowledge of what followed.

7
LIVE TRADES FUNDED
5 closed · 2 open

6 of 7
PROFITABLE
including open positions at mark

+$3,920
LIVE OVERLAY P&L
on $10,000 slices

+1.7 pp
BOOK VS SPY
+4.7% vs +3.0% over 56 sessions

SECURITY	SIGNAL	FILLED	EXIT	ENTRY	EXIT / MARK	P&L	RETURN	EXIT REASON
Pfizer PFE	2026-06-24	2026-06-25	2026-07-28	$23.26	$25.25	+$843	+8.5%	target hit
Verizon VZ	2026-06-30	2026-07-01	2026-07-24	$41.29	$46.38	+$1,223	+12.3%	target hit
AT&T T	2026-06-30	2026-07-01	2026-07-22	$20.21	$23.04	+$1,389	+14.0%	target hit
Intuitive Surgical ISRG	2026-07-17	2026-07-20	2026-08-31	$353.17	$376.86	+$660	+6.7%	time stop
Tesla TSLA	2026-07-23	2026-07-24	2026-07-29	$313.03	$298.32	−$479	−4.7%	divest signal
T-Mobile US TMUS	2026-07-23	2026-07-24	open	$179.06	$182.16	+$163	+1.7%	still open
Linde LIN	2026-07-31	2026-08-03	open	$480.46	$486.76	+$121	+1.3%	still open

The live signal ledger holds 62 signals of all types over the period, of which the washout entries above were the only ones to invest capital. Two things should be read from this table. First, the mechanics work as documented: fills occur at the close after the signal, exits follow the frozen target or the time stop, and the one losing trade was cut by a bearish divest signal rather than allowed to run. Second, 7 trades over 56 sessions is not statistical evidence of edge; it is evidence that the live machine behaves like the walk-forward machine, which is the specific claim the pilot is designed to test at scale.

PROVENANCE SPLIT OF THE FULL RECORD

PROVENANCE	MEANING	SIGNALS	TRADES FUNDED	OVERLAY P&L
Walk-forward	Simulated on history by the causal engine; detected_at = bar date	580	65	+$13,081
Live	Produced by a scheduled run on a new bar; detected_at = run date	62	7	+$3,920

7 Trade analytics

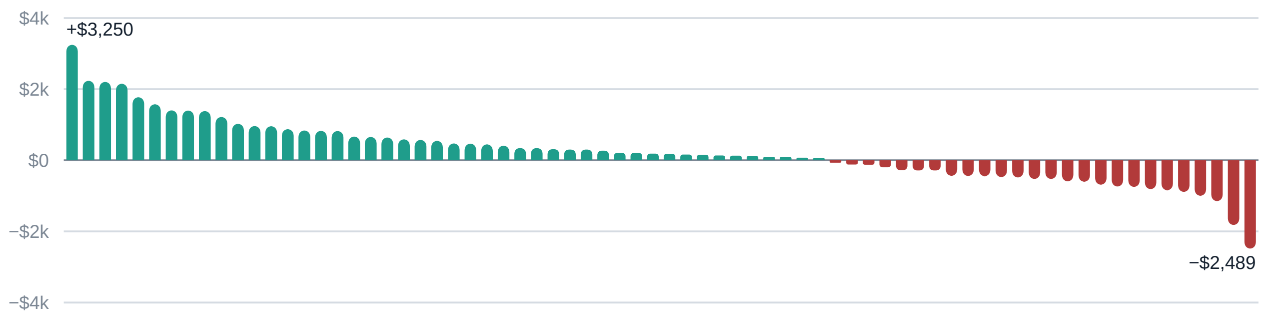
72
ROUND TRIPS
70 closed · 2 open · 45 names

63%
WIN RATE, CLOSED
median trade +2.1%

1.97
PROFIT FACTOR
avg win +$770 · avg loss −$660

32d
AVERAGE HOLD, CALENDAR
up to 13 names held at once

EVERY FUNDED TRADE, RANKED



Dollar profit or loss of each of the 72 slices, sorted. The overlay's +$17,001 is the sum of these bars (+$16,717 realised, +$284 unrealised); the balance of the book's gain, +$33,481, is the SPY baseline itself.

How trades ended

EXIT	TRADES	WINNERS	WIN RATE	AVG RETURN	AVG P&L	TOTAL P&L
Trend target reached	21	21	100%	+10.7%	+$1,046	+$21,968
30-bar time stop	39	23	59%	+1.1%	+$94	+$3,647
Bearish signal — divested	10	0	0%	−9.1%	−$890	−$8,898
Still open	2	2	100%	+1.5%	+$142	+$284

The exit table is the clearest description of the strategy's character. Every trade that reached its frozen trend target was profitable by construction, and those exits carried the bulk of the return. Trades closed by a bearish signal were uniformly losses: the divest rule is the strategy's stop-loss, and the table shows it doing that job at a bounded average cost. The time stop is the mixed bucket: 59% of the 39 time-stopped slices were winners, at a small average result of +$94. The strategy therefore earns its excess return from the tail of large recoveries and pays for it with a larger number of small, capped outcomes.

BY NUMBER OF METHODS IN AGREEMENT

METHODS	CLOSED	WIN RATE	AVG P&L
2 of 4	20	65%	+$310
3 of 4	39	64%	+$238
4 of 4	11	55%	+$111

More agreement did not mean better trades on this sample; the two-method minimum is doing the work. Sub-samples are small and should not be over-read.

BY CATEGORY

CATEGORY	TRADES	WINS	TOTAL P&L
The Mega-Cap 100	36	23	+$14,538
Healthcare & Pharma	3	3	+$2,973
UBS Funds	5	4	+$2,307
Engines of the Republic	5	4	+$1,815
Wall Street's Darlings	2	2	+$273
Big Tech	3	2	+$134
The Clergy House	1	1	+$100
Consumer Leaders	5	2	−$791
Financial Titans	2	1	−$946
Critical Choke Points	10	4	−$3,400

LARGEST WINNERS

SECURITY	FILLED	P&L	RETURN
DELL Dell Technologies	2025-04-04	+$3,250	+32.6%
LLY Eli Lilly	2025-08-08	+$2,237	+22.5%
DIS Walt Disney	2025-04-04	+$2,210	+22.2%
CRWD CrowdStrike	2026-02-24	+$2,155	+21.7%
NVO Novo Nordisk	2025-07-31	+$1,777	+17.9%

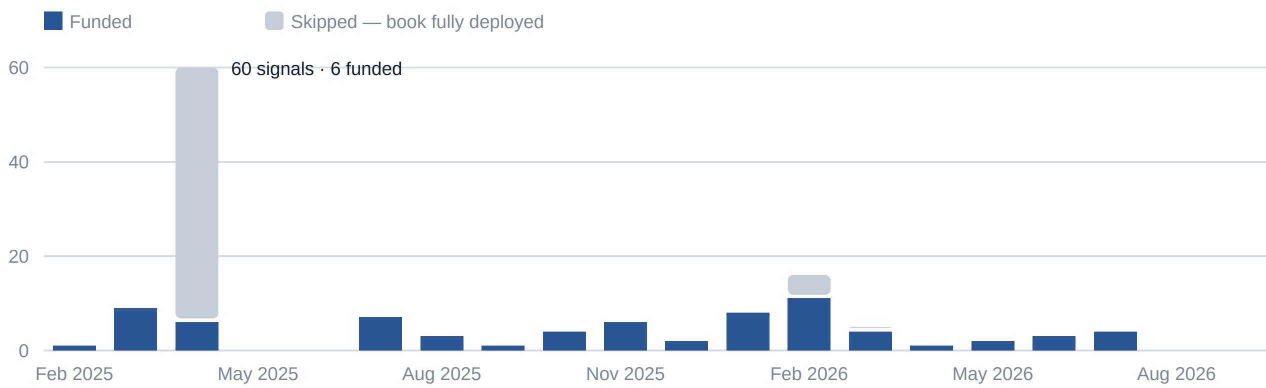
LARGEST LOSERS

SECURITY	FILLED	P&L	RETURN
UNH UnitedHealth Group	2025-04-21	−$2,489	−27.6%
BAC Bank of America	2025-03-05	−$1,825	−18.2%
CRWD CrowdStrike	2026-02-06	−$1,151	−11.4%
DELL Dell Technologies	2025-03-07	−$1,001	−9.9%
DIS Walt Disney	2025-03-12	−$895	−8.9%

The full ledger of 72 trades, with entry and exit prices, exit reasons and the methods that flagged each signal, is Appendix A.

8 Capacity and scaling

The $100,000 book was too small for its own signals. Of 132 eligible washout signals on the record, 72 were funded and 60 (45%) were skipped because every slice was already deployed. This is the single most important fact for a capital decision.



Washout (Buy) signals per month, split into those the book funded and those it had to skip. In Apr 2025 the engine produced 60 washout signals and the book could fund 6.

Washout signals cluster in exactly the episodes the strategy is designed for. A ten-slice book is exhausted within days of a broad sell-off, and every subsequent signal in the cluster goes unfunded until a slice returns. The skipped signals are not a random sample of the record: they are concentrated in the sell-off months. In Apr 2025 the 6 slices the book could fund returned +$5,515 in aggregate, the best monthly cohort on the record, while 54 further signals went unfunded. A larger book is therefore expected to capture more of the effect rather than merely more of the same, subject to the study below.

Two ways to scale

Proportional (validated)
The slice stays at 10% of capital. The book behaves exactly as the record shows, scaled linearly; concentration, drawdown and turnover are unchanged. Skipped signals remain skipped.

Fixed slice (untested)
The slice stays near $10,000 and the additional capital funds signals the small book skipped. More names held at once; single-name weight falls; the equity curve has not been simulated at this configuration.

PRO FORMA ILLUSTRATION, PROPORTIONAL SCALING

TIER	CAPITAL	SLICE	ENDING VALUE	SPY ALONE	OVERLAY CONTRIBUTION	DEEPEST DRAWDOWN
Pilot	$250,000	$25,000	$376,205	$340,357	+$35,848	−$49,850

TIER	CAPITAL	SLICE	ENDING VALUE	SPY ALONE	OVERLAY CONTRIBUTION	DEEPEST DRAWDOWN
Core	$500,000	$50,000	$752,410	$680,714	+$71,696	−$99,701
Scale	$1,000,000	$100,000	$1,504,820	$1,361,427	+$143,392	−$199,402

Hypothetical. The table multiplies the $100,000 record by the capital ratio. It assumes the same 72 trades at the same prices with no market impact, which is reasonable for slices of this size in mega-cap names but is an assumption, not a result. It is an illustration of scale, not a forecast.

> **CONDITION PRECEDENT**
>
> Before any capital is committed, the walk-forward backtest is to be re-run at the proposed capital and slice size under both policies using `reports/prospectus/capacity_study.py`, which drives the production backtest engine with the same ledger and the same fills. The fixed-slice configuration is funded only if that study shows excess return positive in both halves of the window, the acceptance bar already applied to the entry-set decision (Section 10).

9 Risk factors

An allocation to the Signal is an allocation to US large-cap equities with an active overlay. The following factors are material and are stated in the order a treasury committee is likely to weigh them.

Simulated performance and a short live record

The overlay has been active for 20 months in simulation and 56 sessions live. The record spans a single macro regime, a sharp sell-off in early 2025 followed by a strong recovery, which is a favourable environment for washout buying. Reversal strategies can underperform for extended periods, particularly in a persistent bear market where oversold names keep falling; the March 2025 cohort, the worst on the record, is the small-scale example. There is no assurance that the live record will resemble the walk-forward record.

Full equity market risk

The book is always approximately 100% invested in US equities. Its beta to SPY over the record was 1.05 and its deepest drawdown (−18.6%) was the index's own. The overlay does not hedge, does not raise cash in a decline, and cannot protect capital in a market fall. The company should size the allocation as it would size an index position.

Concentration and clustering

Signals cluster in sell-offs, so the book is most exposed to single names precisely when markets are most stressed. Up to 13 names were held at once; each slice is 10% of capital. The universe is tilted toward technology by construction of the mega-cap set. Individual losses can be large: the largest single loss on the record was a −27.6% decline in a slice closed by a divest signal.

Execution assumptions

The simulation fills at the official close of the session after the signal and charges 5 basis points per side. Live execution with market-on-close orders in mega-cap names should approximate this, but slippage, partial fills, halts and the operator's own latency are not modelled. Dividends enter only through adjusted closes. Three names in the universe are open-end mutual funds (DVRUX, QGRPX, BNUEX), which settle at net asset value and commonly carry short-term redemption fees or frequent-trading restrictions; 5 funded trades on the record were in these funds. The pilot should either exclude them or confirm terms with the fund sponsor.

Data dependency

All prices come from one free vendor via `yfinance`. Vendor outages, retroactive adjustments and symbol changes are real; the system mitigates them with retry, a coverage report, a stale-feed watchdog that quarantines flat-lined series, and price-basis reconciliation across corporate actions (3 names carry a basis break today: CRWD, HON, KLAC). A professional data feed is a reasonable upgrade at the Core tier.

Model and parameter risk

Thresholds, weights, the 30-session time stop and the washout-only entry set were chosen by the model owner using this universe and this window. The split-half study is a robustness check, not a fresh out-of-sample test. Parameters are fixed and versioned, and the proposal treats any change to them as an event that re-opens validation (Section 11), but the risk that the rules are tuned to the past cannot be eliminated by any test on the past.

Capacity

Only the $100,000 / $10,000 configuration has been simulated. Scaling the slice with capital preserves the record's behaviour; scaling the number of slices does not, and is untested until the capacity study is run.

Operational and key-person risk

The pipeline runs on GitHub-hosted infrastructure on a schedule; a failed or late run means no signal for that session. One individual currently maintains the code, the universe and the parameters. The proposal assigns an execution owner separate from the model owner and requires that the ledger and the daily health report, not the model owner's account of them, be the basis of oversight.

Tax, accounting and policy

Average holding periods of about a month mean gains will be short-term. Corporate treasury policy, liquidity needs and any covenant restrictions on marketable-securities holdings should be confirmed before funding. Nothing in this document is tax or legal advice.

10 Validation, controls and infrastructure

The causality invariant

Every score computed for a bar uses only that bar and earlier ones. This is not a modelling convention but a test: `test_causality.py` appends future data to a series and fails if any historical verdict changes, for every detector at every sensitivity. The test suite runs before every scheduled detection run. The consequence is that the walk-forward backfill and live operation are literally the same code path, so the historical record is a legitimate out-of-sample-style simulation rather than a fit.

Fixed thresholds and the materiality gate

Signals fire on fixed sigma cut-offs of each security's own standardised deviation. Nothing is a percentile of the run's own output, so a quiet market produces a quiet report and the number of signals is not forced. A trade call needs two methods in agreement and a move of at least 1.0%, which keeps statistically real but economically trivial anomalies in money-market funds out of the trade ledger.

The split-half study that fixed the entry set

Four candidate configurations were run through the production engine on the complete ledger. The acceptance bar was that excess return over the baseline be positive in *both* halves of the window. Adding momentum (Long) entries roughly doubled the trade count, added no excess return, and turned the first half negative; washout-only entries passed under both baselines. The Long signal remains an informational flag.

VARIANT	TRADES	STRATEGY	BASELINE	EXCESS	FIRST HALF	SECOND HALF	VERDICT
BUY + LONG entries, SPY baseline	118	+41.5%	+29.6%	+11.9 pp	−0.7 pp	+9.8 pp	fails first half
BUY-only entries, SPY baseline	65	+41.1%	+29.6%	+11.5 pp	+7.2 pp	+1.7 pp	**accepted**
BUY + LONG entries, BIL baseline	112	+18.6%	+6.5%	+12.1 pp	−1.7 pp	+13.0 pp	fails first half
BUY-only entries, BIL baseline	63	+19.5%	+6.5%	+12.9 pp	+9.6 pp	+2.5 pp	passes (reference)

Source: `HOW_IT_WORKS.md`, regenerated whenever the rules change. The study predates the current run; trade counts have since grown with the live record. The BIL rows show the cash-parked view reaches the same entry-set decision.

Data integrity

- **Price-basis reconciliation.** The vendor rescales an entire history after a split; ledger rows keep the basis of the fetch that wrote them. Each run compares frozen closes with the current fetch per bar, translates frozen trade targets into the current basis exactly, and flags affected names. The ledger itself is never rewritten.
- **Stale-feed watchdog.** A series printing identical closes for five sessions is treated as a dead feed: its bars stay out of the frozen record until it moves again, so a vendor outage cannot mint permanent verdicts.
- **Proof of run.** Every run stamps bars scored and the latest bar date, so a day with no signals is verifiably a quiet market rather than a job that failed. Coverage on the latest run: 100% of 130 securities in 4.3 minutes.

The audit trail

The durable record is two append-only files under `data/ledger/`, committed by every run: 642 signal rows and 1,944 anomaly verdicts today, across 59,537 scored bar-observations. Each row carries the bar date, the date the verdict existed, its provenance and the model version. Because the files are in version control, every verdict ever recorded is a visible line in a commit; an auditor can reconstruct the backtest from the repository alone. The operational dashboard is generated from the same files and published after each run.

Infrastructure

A GitHub Actions workflow runs every weekday at 22:00 UTC, after the US close. It installs dependencies, runs the test suite including the causality invariant, executes the seven-stage pipeline, commits the ledger, health report and run history, and deploys the dashboard. There are no servers to maintain and no credentials beyond the repository’s own. The run is idempotent: a re-run on the same day writes nothing new.

11 Proposed allocation of company capital

A staged programme that funds the validated structure first, measures the live record against explicit guardrails, and scales only on evidence.

Tiers

TIER	CAPITAL	SLICE	SIZING POLICY	ENTRY GATE
I · Pilot	$250,000	$25,000	Proportional; ten-slice book identical in behaviour to the record	Conditions precedent below
II · Core	$500,000	$50,000	Proportional, or fixed-slice if the capacity study passes	90-day review gate passed
III · Scale	$1,000,000	$50,000–$100,000	Per the capacity study; professional data feed in place	Second review gate passed; 12 months live

Conditions precedent to funding Tier I

- Board or treasury-committee approval of the allocation and of this document’s guardrails as policy.
- A segregated brokerage sub-account funded in SPY, with authority limited to the universe in Appendix B, long-only, no margin, no options.
- The capacity study run at $250,000 / $25,000, confirming the proportional configuration reproduces the record.
- An execution owner named, distinct from the model owner, with a written daily procedure: read the dashboard after the 22:00 UTC run, place market-on-close orders for that day’s entries and exits on the next session.
- Mutual funds in the universe either excluded from live trading or cleared with the sponsor.
- Baseline reporting in place: the daily dashboard link, and this document regenerated monthly.

Implementation timeline

1. **Weeks 0–2 · Approval and set-up.** Policy amendment, account opening, execution procedure, capacity study, mutual-fund decision.
2. **Months 1–3 · Pilot.** Tier I funded. Daily execution. Monthly one-page memo generated from the ledger. No parameter changes.
3. **Month 3 · First review gate.** Live record measured against the guardrails. Proceed to Tier II, hold, or wind down.

4. **Months 4–12 · Core.** Tier II funded. Quarterly review. Data-feed upgrade evaluated.
5. **Month 12 · Second review gate.** Twelve months of live record. Tier III decision.

Guardrails

Each rule is measured from the committed ledger and the daily portfolio value, not from anyone's judgement, and each names its consequence.

TRIGGER	MEASURED AS	CONSEQUENCE
Book drawdown exceeds 20% from peak	Daily portfolio value	No new entries; open slices run to their exits; committee review before resumption
Overlay lags SPY by more than 8 percentage points over a trailing 6 months	Book value vs baseline value	Halt new entries; model owner reports; committee decides
Trailing 30 closed live trades: profit factor below 1.0 or win rate below 45%	Signals ledger, live provenance	Halt new entries pending review
Fetch coverage below 95% on three consecutive runs, or a stale-feed flag on a held name	`run_health.json`	No new entries until coverage recovers; held names exit by rule
Any change to parameters, thresholds, universe membership or model version	Repository diff	Trading pauses until the split-half study and capacity study are re-run and approved
Scheduled run fails or is late	Workflow status	No trades that session; exits by rule on the next successful run

Governance and reporting

ROLE	HOLDER	RESPONSIBILITY
Allocation owner	Chief Executive / treasury committee	Approves tiers, gates and any resumption after a guardrail
Model owner	Jacob E. Thomas, PhD	Maintains code, universe and parameters; may not execute trades
Execution owner	Treasury / finance designee	Places and confirms orders; reconciles fills against the ledger
Oversight	Finance or internal audit	Reads the ledger and health report directly; certifies the monthly memo

Reporting cadence. Daily: the dashboard, published automatically. Monthly: this document regenerated from the ledger, with the live section current. Quarterly: committee review of guardrails, capacity and data quality. Ad hoc: any guardrail breach, reported the same day.

12 Summary of key terms

Strategy	The Signal — systematic washout-reversion overlay on an S&P 500 position
Manager / model owner	Jacob E. Thomas, PhD, Results Generation
Vehicle	Segregated internal brokerage sub-account; no external investors, no fees
Objective	Outperform SPY held passively, with the same market exposure
Base currency	US dollars
Baseline asset	SPY (SPDR S&P 500 ETF); all idle capital
Universe	130 US-listed securities (Appendix B); registry in code
Entry rule	Buy signal: ≤ −2.5σ from 20-day trend, 2+ methods agree, |move| ≥ 1.0%; fill at next session's close
Exit rules	Frozen trend target, first bearish signal on the name, or 30-session time stop, whichever first
Position size	10% of capital per slice; maximum ten concurrent slices at inception capital
Leverage / shorting / derivatives	None / none / none
Liquidity	Daily; every holding is an exchange-listed security or a daily-NAV fund
Valuation	Daily at official closes; ledger and dashboard refreshed each weekday at 22:00 UTC
Record	Walk-forward from 2024-11-01; live since 2026-06-12; model version 2.0.0
Performance to date	\$100,000 → \$150,482 (+50.5%) vs \$136,143 (+36.1%) in SPY; Sharpe 1.28; max drawdown −18.6%
Proposed capital	\$250,000 pilot → \$500,000 → \$1,000,000, gated (Section 11)
Guardrails	Drawdown, relative-performance, hit-rate, data-coverage and change-control halts (Section 11)
Reporting	Daily dashboard; monthly regenerated prospectus; quarterly committee review
Audit trail	Append-only ledger in version control; every verdict reproducible from the repository
Data	Yahoo Finance via `yfinance`; coverage and health reported per run

A Complete trade ledger

All 72 funded slices in order of fill, exactly as the walk-forward backtest records them. Provenance: W = walk-forward, L = live. Prices are in the current adjusted basis; CRWD rows were basis-translated across a 4:1 split. Methods list the detectors that flagged the signal bar.

#	TICKER	P	SIGNAL	FILLED	EXIT	ENTRY	EXIT / MARK	P&L	RETURN	EXIT REASON	METHODS
1	MRK	B	2025-02-04	2025-02-05	2025-03-20	$85.17	$90.76	+$645	+6.6%	time stop	Ensemble, EWMA
2	BAC	B	2025-03-04	2025-03-05	2025-04-04	$40.89	$33.46	−$1,825	−18.2%	divest signal	Fourier, Matrix profile, Ensemble, EWMA
3	DELL	B	2025-03-06	2025-03-07	2025-04-21	$89.37	$80.51	−$1,001	−9.9%	time stop	Fourier, Ensemble, EWMA
4	QGRPX	B	2025-03-10	2025-03-11	2025-04-23	$15.13	$14.46	−$451	−4.4%	time stop	Fourier, Ensemble, EWMA
5	ISRG	B	2025-03-10	2025-03-11	2025-04-23	$493.72	$487.93	−$127	−1.2%	time stop	Ensemble, EWMA
6	AXP	B	2025-03-10	2025-03-11	2025-04-23	$251.22	$256.74	+$210	+2.2%	time stop	Fourier, Ensemble, EWMA
7	RY	B	2025-03-11	2025-03-12	2025-03-25	$107.95	$113.16	+$472	+4.8%	target hit	Fourier, Ensemble, EWMA
8	DIS	B	2025-03-11	2025-03-12	2025-04-24	$96.94	$88.35	−$895	−8.9%	time stop	Fourier, Ensemble, EWMA
9	TGT	B	2025-03-12	2025-03-13	2025-04-04	$98.23	$89.98	−$849	−8.4%	divest signal	Fourier, Ensemble, EWMA
10	TGT	B	2025-03-13	2025-03-14	2025-04-04	$98.42	$89.98	−$818	−8.6%	divest signal	Fourier, Ensemble, EWMA
11	QGRPX	B	2025-04-03	2025-04-04	2025-05-02	$13.65	$15.58	+$1,401	+14.1%	target hit	Fourier, Ensemble, EWMA
12	NKE	B	2025-04-03	2025-04-04	2025-05-19	$55.35	$60.02	+$833	+8.4%	time stop	Fourier, Ensemble, EWMA
13	DIS	B	2025-04-03	2025-04-04	2025-05-07	$81.99	$100.21	+$2,210	+22.2%	target hit	Fourier, Ensemble, EWMA

#	TICKER	P	SIGNAL	FILLED	EXIT	ENTRY	EXIT / MARK	P&L	RETURN	EXIT REASON	METHODS
14	DELL	B	2025-04-03	2025-04-04	2025-04-24	$69.99	$92.83	+$3,250	+32.6%	target hit	Matrix profile, Ensemble, EWMA
15	AAPL	B	2025-04-08	2025-04-09	2025-04-29	$197.63	$209.92	+$310	+6.2%	target hit	Fourier, Ensemble, EWMA
16	UNH	B	2025-04-17	2025-04-21	2025-05-14	$411.32	$297.86	−$2,489	−27.6%	divest signal	Fourier, Ensemble, EWMA
17	ABT	B	2025-07-17	2025-07-18	2025-08-07	$120.88	$129.08	+$668	+6.8%	target hit	Fourier, Matrix profile, Ensemble, EWMA
18	LMT	B	2025-07-22	2025-07-23	2025-09-04	$408.38	$448.10	+$962	+9.7%	time stop	Fourier, Ensemble, EWMA
19	TXN	B	2025-07-23	2025-07-24	2025-09-05	$179.70	$183.19	+$184	+1.9%	time stop	Ensemble, EWMA
20	STM	B	2025-07-24	2025-07-25	2025-09-08	$26.04	$26.33	+$100	+1.1%	time stop	Fourier, Matrix profile, Ensemble, EWMA
21	NVO	B	2025-07-29	2025-07-30	2025-09-11	$47.12	$51.72	+$965	+9.8%	time stop	Matrix profile, Ensemble, EWMA
22	NVO	B	2025-07-30	2025-07-31	2025-09-12	$44.33	$52.26	+$1,777	+17.9%	time stop	Fourier, Matrix profile, Ensemble, EWMA
23	PANW	B	2025-07-31	2025-08-01	2025-09-08	$172.88	$197.38	+$1,406	+14.2%	target hit	Ensemble, EWMA
24	BRK-B	B	2025-08-04	2025-08-05	2025-08-13	$464.19	$477.31	+$272	+2.8%	target hit	Matrix profile, Ensemble, EWMA
25	MCK	B	2025-08-07	2025-08-08	2025-08-21	$665.87	$703.14	+$549	+5.6%	target hit	Fourier, Ensemble, EWMA
26	LLY	B	2025-08-07	2025-08-08	2025-09-16	$620.37	$759.92	+$2,237	+22.5%	target hit	Ensemble, EWMA
27	ISRG	B	2025-09-03	2025-09-04	2025-10-16	$454.52	$434.62	−$447	−4.4%	time stop	Fourier, Matrix profile, Ensemble, EWMA
28	T	B	2025-10-06	2025-10-07	2025-11-18	$24.98	$24.71	−$119	−1.1%	time stop	Ensemble, EWMA

#	TICKER	P	SIGNAL	FILLED	EXIT	ENTRY	EXIT / MARK	P&L	RETURN	EXIT REASON	METHODS
29	XLF	B	2025-10-16	2025-10-17	2025-10-27	$51.55	$52.69	+$210	+2.2%	target hit	Fourier, Matrix profile, Ensemble, EWMA
30	META	B	2025-10-31	2025-11-03	2025-12-16	$636.06	$655.98	+$303	+3.1%	time stop	Fourier, Matrix profile, Ensemble, EWMA
31	LIN	B	2025-10-31	2025-11-03	2025-12-16	$408.00	$420.76	+$302	+3.1%	time stop	Ensemble, EWMA
32	LIN	B	2025-11-03	2025-11-04	2025-12-17	$413.70	$419.60	+$132	+1.4%	time stop	Ensemble, EWMA
33	HD	B	2025-11-18	2025-11-19	2026-01-05	$327.69	$339.26	+$343	+3.5%	time stop	Fourier, Ensemble, EWMA
34	UBER	B	2025-11-20	2025-11-21	2025-12-08	$83.87	$92.57	+$1,026	+10.4%	target hit	Fourier, Ensemble, EWMA
35	PANW	B	2025-11-20	2025-11-21	2026-01-07	$182.90	$193.90	+$591	+6.0%	time stop	Ensemble, EWMA
36	ETN	B	2025-11-20	2025-11-21	2026-01-07	$328.96	$319.99	−$282	−2.7%	time stop	Fourier, Ensemble, EWMA
37	BNUEX	B	2025-11-20	2025-11-21	2025-11-25	$11.91	$12.15	+$187	+2.0%	target hit	Fourier, Ensemble, EWMA
38	PG	B	2025-12-08	2025-12-09	2025-12-17	$136.59	$144.59	+$575	+5.9%	target hit	Fourier, Ensemble, EWMA
39	AVGO	B	2025-12-15	2025-12-16	2026-01-30	$339.40	$330.08	−$284	−2.7%	time stop	Fourier, Ensemble, EWMA
40	MSFT	B	2026-01-21	2026-01-22	2026-01-26	$448.30	$467.32	+$414	+4.2%	target hit	Fourier, Ensemble, EWMA
41	ABT	B	2026-01-22	2026-01-23	2026-03-09	$105.99	$111.15	+$476	+4.9%	time stop	Fourier, Ensemble, EWMA
42	ABT	B	2026-01-23	2026-01-26	2026-03-10	$107.33	$109.08	+$153	+1.6%	time stop	Fourier, Ensemble, EWMA
43	UNH	B	2026-01-27	2026-01-28	2026-01-29	$290.10	$288.39	−$69	−0.6%	divest signal	Fourier, Matrix profile, Ensemble, EWMA

#	TICKER	P	SIGNAL	FILLED	EXIT	ENTRY	EXIT / MARK	P&L	RETURN	EXIT REASON	METHODS
44	SAP	B	2026-01-29	2026-01-30	2026-02-03	$197.60	$193.92	−$196	−1.9%	divest signal	Fourier, Matrix profile, Ensemble, EWMA
45	MSFT	B	2026-01-29	2026-01-30	2026-03-16	$427.58	$398.34	−$693	−6.8%	time stop	Fourier, Matrix profile, Ensemble, EWMA
46	UNH	B	2026-01-30	2026-02-02	2026-03-17	$281.78	$285.94	+$137	+1.5%	time stop	Fourier, Ensemble, EWMA
47	SAP	B	2026-01-30	2026-02-02	2026-02-03	$203.69	$193.92	−$489	−4.8%	divest signal	Fourier, Ensemble, EWMA
48	UNH	B	2026-02-02	2026-02-03	2026-03-18	$280.39	$282.72	+$73	+0.8%	time stop	Fourier, Ensemble, EWMA
49	SAP	B	2026-02-03	2026-02-04	2026-03-19	$196.27	$181.85	−$744	−7.3%	time stop	Fourier, Ensemble, EWMA
50	MSFT	B	2026-02-03	2026-02-04	2026-03-19	$411.58	$387.45	−$596	−5.9%	time stop	Fourier, Ensemble, EWMA
51	PLTR	B	2026-02-04	2026-02-05	2026-03-20	$130.01	$150.68	+$1,578	+15.9%	time stop	Fourier, Ensemble, EWMA
52	MSFT	B	2026-02-04	2026-02-05	2026-03-20	$391.19	$380.33	−$287	−2.8%	time stop	Fourier, Ensemble, EWMA
53	PANW	B	2026-02-05	2026-02-06	2026-02-19	$159.32	$150.99	−$532	−5.2%	divest signal	Ensemble, EWMA
54	MSFT	B	2026-02-05	2026-02-06	2026-03-23	$398.61	$381.45	−$440	−4.3%	time stop	Fourier, Ensemble, EWMA
55	CRWD	B	2026-02-05	2026-02-06	2026-02-23	$98.88	$87.58	−$1,151	−11.4%	divest signal	Ensemble, EWMA
56	ARKK	B	2026-02-05	2026-02-06	2026-03-23	$70.41	$70.92	+$62	+0.7%	time stop	Fourier, Ensemble, EWMA
57	IBM	B	2026-02-23	2026-02-24	2026-04-08	$226.02	$238.26	+$450	+5.4%	time stop	Ensemble, EWMA
58	CRWD	B	2026-02-23	2026-02-24	2026-03-05	$87.56	$106.54	+$2,155	+21.7%	target hit	Ensemble, EWMA
59	BLK	B	2026-03-06	2026-03-09	2026-04-21	$952.30	$1,037.01	+$879	+8.9%	time stop	Fourier, Ensemble, EWMA
60	DVRUX	B	2026-03-12	2026-03-13	2026-04-09	$16.89	$17.49	+$345	+3.5%	target hit	Ensemble, EWMA

#	TICKER	P	SIGNAL	FILLED	EXIT	ENTRY	EXIT / MARK	P&L	RETURN	EXIT REASON	METHODS
61	ABBV	B	2026-03-18	2026-03-19	2026-05-01	$203.09	$205.14	+$91	+1.0%	time stop	Ensemble, EWMA
62	QGRPX	B	2026-03-27	2026-03-30	2026-04-08	$16.05	$17.39	+$824	+8.3%	target hit	Ensemble, EWMA
63	NKE	B	2026-04-01	2026-04-02	2026-05-15	$43.80	$41.51	−$532	−5.2%	time stop	Ensemble, EWMA
64	ISRG	B	2026-05-11	2026-05-12	2026-06-25	$431.87	$399.69	−$754	−7.5%	time stop	Ensemble, EWMA
65	WMT	B	2026-05-21	2026-05-22	2026-07-08	$119.98	$112.83	−$606	−6.0%	time stop	Ensemble, EWMA
66	PFE	L	2026-06-24	2026-06-25	2026-07-28	$23.26	$25.25	+$843	+8.5%	target hit	Fourier, Ensemble, EWMA
67	VZ	L	2026-06-30	2026-07-01	2026-07-24	$41.29	$46.38	+$1,223	+12.3%	target hit	Matrix profile, Ensemble, EWMA
68	T	L	2026-06-30	2026-07-01	2026-07-22	$20.21	$23.04	+$1,389	+14.0%	target hit	Fourier, Matrix profile, Ensemble, EWMA
69	ISRG	L	2026-07-17	2026-07-20	2026-08-31	$353.17	$376.86	+$660	+6.7%	time stop	Ensemble, EWMA
70	TSLA	L	2026-07-23	2026-07-24	2026-07-29	$313.03	$298.32	−$479	−4.7%	divest signal	Fourier, Ensemble, EWMA
71	TMUS	L	2026-07-23	2026-07-24	open	$179.06	$182.16	+$163	+1.7%	still open	Ensemble, EWMA
72	LIN	L	2026-07-31	2026-08-03	open	$480.46	$486.76	+$121	+1.3%	still open	Matrix profile, Ensemble, EWMA

B The universe in full

ENGINES OF THE REPUBLIC · 15 names

CAT Caterpillar · DE Deere & Company · HON Honeywell · LMT Lockheed Martin · GE GE Aerospace · WMT Walmart · COST Costco · HD Home Depot · JNJ Johnson & Johnson · PFE Pfizer · ADM Archer-Daniels-Midland · BG Bunge Global · TSN Tyson Foods · SCHW Charles Schwab · BRK-B Berkshire Hathaway B

CRITICAL CHOKE POINTS · 14 names

NEE NextEra Energy · D Dominion Energy · KMI Kinder Morgan · WMB Williams Companies · V Visa · MA Mastercard · JPM JPMorgan Chase · MSFT Microsoft · AMZN Amazon · GOOGL Alphabet · UNP Union Pacific · NSC Norfolk Southern · MCK McKesson · UNH UnitedHealth Group

BIG TECH · 8 names

AAPL Apple · NVDA NVIDIA · META Meta Platforms · TSLA Tesla · NFLX Netflix · CRM Salesforce · AMD AMD · ORCL Oracle

CONSUMER LEADERS · 8 names

PG Procter & Gamble · KO Coca-Cola · PEP PepsiCo · MCD McDonald's · NKE Nike · SBUX Starbucks · TGT Target · LOW Lowe's

FINANCIAL TITANS · 6 names

GS Goldman Sachs · MS Morgan Stanley · BAC Bank of America · C Citigroup · WFC Wells Fargo · BLK BlackRock

HEALTHCARE & PHARMA · 5 names

LLY Eli Lilly · ABBV AbbVie · MRK Merck · AMGN Amgen · BMY Bristol-Myers Squibb

ENERGY & COMMODITIES · 3 names

XOM ExxonMobil · CVX Chevron · COP ConocoPhillips

WALL STREET'S DARLINGS · 10 names

SPY SPDR S&P 500 ETF · QQQ Invesco QQQ (Nasdaq 100) · DIA SPDR Dow Jones ETF · IWM iShares Russell 2000 · VTI Vanguard Total Stock Market · VOO Vanguard S&P 500 · ARKK ARK Innovation ETF · VGT Vanguard Info Tech ETF · XLF Financial Select SPDR · XLE Energy Select SPDR

THE MEGA-CAP 100 · 55 names

TSM Taiwan Semiconductor (ADR) · AVGO Broadcom · MU Micron Technology · ASML ASML Holding (ADR) · INTC Intel · CSCO Cisco Systems · LRCX Lam Research · AMAT Applied Materials · ARM Arm Holdings (ADR) · PLTR Palantir · HSBC HSBC Holdings (ADR) · PM Philip Morris · NVS Novartis (ADR) · KLAC KLA Corporation · AZN AstraZeneca (ADR) · BABA Alibaba (ADR) · RY Royal Bank of Canada · TXN Texas Instruments · IBM IBM · DELL Dell Technologies · RTX RTX Corporation · SHEL Shell (ADR) · LIN Linde · GEV GE Vernova · TM Toyota Motor (ADR) · MRVL Marvell Technology · PANW Palo Alto Networks · AXP American Express · QCOM Qualcomm · TMUS T-Mobile US · SAP SAP (ADR) · TTE TotalEnergies (ADR) · VZ Verizon · ADI Analog Devices · ANET Arista Networks · NVO Novo Nordisk (ADR) · TD Toronto-Dominion Bank · TJX TJX Companies · APH Amphenol · TMO Thermo Fisher Scientific · DIS Walt Disney · APP AppLovin · CRWD CrowdStrike · T AT&T · ABT Abbott Laboratories · GILD Gilead Sciences · WELL Welltower · ISRG Intuitive Surgical · ETN Eaton · BX Blackstone · SHOP Shopify · UBER Uber Technologies · PLD Prologis · DHR Danaher · CB Chubb

THE CLERGY HOUSE · 2 names

ASTS AST SpaceMobile · STM STMicroelectronics

RESERVE · 1 names

BIL SPDR Bloomberg 1-3 Month T-Bill

UBS FUNDS · 3 names

DVRUX UBS US Dividend Ruler P · QGRPX UBS US Quality Growth P · BNUEX UBS Intl Sustainable Equity P

C Methodology notes and glossary

COMPUTATION OF STATISTICS

- **Daily returns** are computed from the daily mark-to-market value of the book (baseline shares at the SPY close plus open slices at their closes), as the backtest records it.
- **CAGR** compounds the total return over the calendar span in years (1.83). **Volatility** is the standard deviation of daily returns × √252.
- **Sharpe** = mean daily return ÷ standard deviation × √252, risk-free rate 0%, matching the dashboard. **Sortino** uses the downside deviation (negative days only, over all days) in the denominator.
- **Maximum drawdown** is the largest peak-to-trough decline of the daily series, divided by the value at that peak. The operational dashboard's drawdown tile divides the same dollar decline by the series' all-time high, which understates the figure (it shows 13.0% where the peak-to-trough measure is 18.6%); this document uses the conventional measure and the dashboard tile should be read accordingly.
- **Beta, correlation and alpha** are from daily returns against SPY; alpha is annualised (× 252). **Tracking error** is the standard deviation of daily active return × √252; the **information ratio** is annualised active return ÷ tracking error. **Capture ratios** sum the book's returns on the index's up (down) days over the index's own.
- **Trade statistics** use closed round trips only, except where marked. **Profit factor** is gross profit ÷ gross loss. Holding periods are calendar days between fill and exit; the dashboard reports trading sessions.
- **Monthly returns** use month-end portfolio values; the first month runs from the anchor date.

GLOSSARY

Anchor date	The fixed start of the analysis window. Detection and the backtest run forward from it; anchoring rather than sliding keeps every historical verdict stable.
Causal	A computation for bar *t* that uses only bars up to and including *t*. Every detector in the system is causal, and the test suite enforces it.
dev_z	Deviation of price from its 20-day exponential moving average, divided by that deviation's own trailing volatility. The unit in which all thresholds are set.
Divest signal	A Sell or Short verdict on a name the book holds. Moves the slice back to SPY; never opens a short.
Frozen target	The trend value recorded at detection. A Buy slice exits when the close touches it; the value is translated across corporate actions but never re-estimated.
Left matrix profile	For each 10-day pattern, the distance to its most similar pattern strictly earlier in the series. It cannot match against the future by construction.

Materiality gate	The minimum absolute distance from trend (1.0%) before a statistical anomaly may become a trade call.
Provenance	Whether a verdict was produced by the walk-forward simulation over history or by a scheduled run on a new bar. Reported separately, always.
Slice	The fixed dollar amount moved from SPY into a security on a Buy signal.
Time stop	The maximum holding period, 30 trading sessions, after which a slice returns to SPY regardless of price.
Walk-forward	A simulation that advances one bar at a time, producing at each step only what a live run that evening would have produced.
Watermark	Per-security high-water mark of scored bars. Only bars beyond it are persisted, so a verdict is written exactly once.

D Data provenance and reproduction

This document was generated on 2026-09-02 17:37 UTC from repository commit `6c66783`, whose most recent scheduled run is dated 2026-09-01 with the latest bar at 2026-09-01. The following files were read; nothing else contributed a number.

`docs/index.html`	The rendered dashboard: daily equity and benchmark series, drawdown series and the trade ledger of the walk-forward backtest
`data/ledger/signals.jsonl`	The append-only signal ledger (642 rows): signal type, bar date, detection date, provenance, confidence and the frozen rationale
`data/history/run_2026-09-01.json`	Run summary: observations scored, ledger row counts, backtest totals, fetch report
`data/run_health.json`	Coverage, fetch failures, stale feeds, price-basis breaks
`data/alerts.json`	Run block and per-security status for the latest run
`anomaly_detection/config.py`	Universe registry, category labels, protocol parameters
`HOW_IT_WORKS.md`	The split-half robustness study (Section 10), transcribed

TO REPRODUCE

From the repository root, after any scheduled run has committed: `python reports/prospectus/build_prospectus.py --pdf`. The HTML is self-contained apart from web fonts; the PDF is rendered by headless Chromium via `reports/prospectus/render_pdf.cjs`. To reproduce the backtest itself: `python -m anomaly_detection` rebuilds the store from the ledger, re-fetches prices, re-scores, and re-runs the walk-forward simulation; because scoring is causal and the ledger is frozen, the trade list is invariant to when it is run.

SIGNAL LEDGER COMPOSITION

SIGNAL	ROWS
BUY	133
SELL	184
LONG	113
SHORT	169
REDUCE	10
WATCH	33
Total	**642**

Results Generation · Jacob E. Thomas, PhD · The Signal, model version 2.0.0 · Generated 2026-09-02 17:37 UTC · Confidential.